

Genome-wide association study of obstructive sleep apnoea in the Million Veteran Program uncovers genetic heterogeneity by sex



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Summary

Background Genome-wide association studies (GWAS) for obstructive sleep apnoea (OSA) are limited due to the underdiagnosis of OSA, leading to misclassification of OSA, which consequently reduces statistical power. We performed a GWAS of OSA in the Million Veteran Program (MVP) of the U.S. Department of Veterans Affairs (VA) healthcare system, where OSA prevalence is close to its true population prevalence.

Methods We performed GWAS of 568,576 MVP participants, stratified by biological sex and by harmonized race/ethnicity and genetic ancestry (HARE) groups of White, Black, Hispanic, and Asian individuals. We considered both BMI adjusted (BMI-adj) and unadjusted (BMI-unadj) models. We replicated associations in independent datasets, and analysed the heterogeneity of OSA genetic associations across HARE and sex groups. We finally performed a larger meta-analysis GWAS of MVP, FinnGen, and the MGB Biobank, totalling 916,696 individuals.

Findings MVP participants are 91% male. OSA prevalence is 21%. In MVP there were 18 and 6 genome-wide significant loci in BMI-unadj and BMI-adj analyses, respectively, corresponding to 21 association regions. Of these, 17 were not previously reported in association with OSA, and 13 replicated in FinnGen (False Discovery Rate p -value < 0.05). There were widespread significant differences in genetic effects between men and women, but less so across HARE groups. Meta-analysis of MVP, FinnGen, and MGB biobank revealed 17 additional, previously unreported, genome-wide significant regions.

Interpretation Sex differences in genetic associations with OSA are widespread, likely associated with multiple OSA risk factors. OSA shares genetic underpinnings with several sleep phenotypes, suggesting shared aetiology and causal pathways.

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Introduction

Obstructive sleep apnoea (OSA) is a condition characterized by repeated airway collapses during sleep, leading to intermittent reductions in oxyhaemoglobin saturation and cortical arousals during sleep.¹ The heritabilities of

OSA-related phenotypes have been estimated to be up to 25%–40% in family studies,^{2–4} and up to 17% in population-based studies.^{5,6} Genome-wide association studies (GWAS) of OSA and related phenotypes reported a handful of genetic variants,^{6–11} mostly related to

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Research in context

Evidence before this study

Few genome-wide association studies (GWAS) of obstructive sleep apnea (OSA) were reported in the literature. Moreover, they were limited by under-diagnosis of OSA, which limits statistical power. Also, prior studies had not addressed in GWAS the known heterogeneity of OSA by biological sex and other demographic factors.

Added value of this study

GWAS of OSA in the Million Veteran Program (MVP) with and without adjustment to body mass index (BMI) and meta-analysis with additional GWAS detected 32 genomic loci associated with OSA that were not previously known. Analysis of sex differences demonstrates that many genetic variants across the genome have different OSA associations between men and women and the associations of sex-different OSA

variants tend to be protective against OSA in women. Analysis of genetic correlations and pleiotropy suggests that the genetic association of OSA with cardiometabolic phenotypes is mainly driven by BMI, but there is a component of genetic correlation with type 2 diabetes that is independent of BMI.

Implications of all the available evidence

Known sex differences in OSA, with OSA substantially more prevalent in men compared to women, have a genetic basis, which may correspond to the aggregated effects of multiple OSA risk factors. Results from BMI-adjusted OSA GWAS may be used to study genetic components of OSA that are independent of BMI and obesity, and, following additional implementation studies, may help identify individuals at risk of OSA who in whom diagnosis might not be considered due to absence of obesity.

cardiometabolic conditions such as obesity and lung function. There are known sex, gender and race/ethnic differences in OSA,^{12–14} but their genetic basis has not been thoroughly studied, mainly due to sample size and resulting power limitations. Because BMI and other adiposity and body fat distribution measures are strong causal determinants of OSA,¹⁵ it is important to account for it when studying the genetic basis of OSA. Until recently, the largest GWAS of OSA and of a proxy phenotype, snoring, primarily reported results from analyses that did not adjust for BMI.^{6,16} A recent publication made a considerable progress and reported a multi-trait analysis of OSA and snoring, followed with replication analysis of top associations in a BMI-adjusted analysis in 23andMe.¹⁷

Critically for GWAS, OSA is common yet under-diagnosed.^{18,19} In healthcare settings, OSA is typically evaluated by indication, following symptoms such as excessive daytime sleepiness and in individuals who are perceived to be more likely to have OSA, i.e., those who are overweight or obese.²⁰ Thus, individuals with OSA are often not diagnosed. For example, a publication based on health records reported <1% individuals with the diagnosis of OSA,²¹ while another study identified 9% prevalence of moderate-to-severe OSA among individuals without OSA history.²² Estimates from population-based cohort studies in adults, which are not biased by indication for assessment, are 9%–38% for mild to severe OSA and 2%–17% of moderate to severe OSA.^{23,24} Thus, statistical power for genetic analysis of OSA is often limited: while OSA can be studied based on large biobanks, its underdiagnosis limits this investigation, e.g., only 1.47% of UK Biobank individuals have OSA according to the electronic health records.¹⁶ In FinnGen, based on which a GWAS of OSA was recently published, the OSA rate was 8%,⁶ which also likely points to misclassification of many “controls”. In the

23andMe analysis reported by Campos et al.,¹⁷ ~12% participants self-reported OSA. Population-based cohort studies which measure all individuals regardless of indications are available, and were used for GWAS of OSA-related measures, but their sample sizes are limited.^{9–11,25}

In contrast to the prevalence of OSA identified from health records in other healthcare systems, a high prevalence of OSA has been identified in patients receiving care from the U.S. Department of Veterans Affairs (VA) healthcare system, with a reported prevalence of sleep-related breathing disorders of 22.2% in 2018, based on International Classification of Disease (ICD) codes.²⁶ This reflects a 4-fold increase in prevalence from 2012 and follows a 6-fold increase between 2000 and 2010.²⁷ While the prevalence of OSA in the VA patient population is likely to exceed that in the general population, other factors have been largely responsible for the increase in identification of OSA, including the early and widespread adoption of home sleep apnoea testing in the VA, the low cost to Veterans for diagnostic services, the lack of copayments for prosthetic equipment used in OSA treatment, and financial incentives for service-related illnesses providing an incentive for patients with sleep problems to pursue a diagnosis. The prevalence of diagnosed OSA in the VA is therefore much closer to the expected population prevalence of OSA when compared to reports from other healthcare settings, likely resulting in lower misclassification rate (i.e., fewer individuals with OSA being identified as not having OSA in the medical record). Here, data from 568,576 individuals from the VA's Million Veteran Program (MVP) contributed to the GWAS of OSA. Analyses accounted for BMI and stratified by sex and by groups approximating race and ethnicity, while prioritizing genetic similarity. We investigated genetic determinants of OSA and their heterogeneity across these

population subgroups, and genetic relationship to other phenotypes. We also performed a larger meta-analysis GWAS of MVP, FinnGen, and the Mass General Brigham (MGB) Biobank, totalling 916,696 individuals.

Methods

Obstructive sleep apnoea phenotype definition and covariates

All participants were Veterans and OSA was identified from the VA electronic health record using the previously published multimodal automated phenotyping (MAP) procedure (Liao et al., 2019). MAP is an unsupervised clustering algorithm that uses counts of ICD codes from the electronic medical records and concept unique identifiers (CUI) derived by natural language processing of text notes, along with total number of notes as a proxy for healthcare utilization, and provides a predicted probability that a patient has the phenotype of interest and a threshold probability for a binary classification of the phenotype as present or absent. To identify OSA, the MAP procedure utilized ICD-9 code 327.23 and ICD-10 code G47.33, and CUI C0520679 (Sleep Apnea, Obstructive) in the Unified Medical Language System (UMLS) (<https://uts.nlm.nih.gov/uts/umls/concept/C0520679>). To identify sleep apnea more generally (i.e., not restricted to obstructive sleep apnoea), the MAP procedure utilized ICD-9 codes 327.2x, 780.51, 780.53, 780.57, ICD-10 codes G47.3X, and CUIs C1561861, C1561869, C0037315, C0751762, C1561862 in the UMLS. The MAP probability cutoffs were 0.38 for MAP-OSA and 0.37 for MAP-SA. The OSA phenotype for this analysis included all individuals with MAP-OSA ≥ 0.38 plus those with both a MAP-OSA > 0 (implying at least one ICD code for OSA) and MAP-SA ≥ 0.37 , in order to capture patients whose OSA had been coded using a general term for sleep apnoea. This OSA phenotype was validated by reviewing 100 charts randomly selected from patients who had at least one ICD code for OSA and had a positive predictive value of 0.92. Individual level MVP data can be accessed on a “need to know” basis to eligible MVP investigators.

Analyses were adjusted for age at the first OSA ICD code in OSA cases and at the age during last VA encounter in OSA controls, and the first 10 PCs of genetic data. In addition, a BMI adjusted model further adjusted for BMI, using both linear and squared terms. BMI measures were taken from the electronic health records from a visit within a year of the participant's first OSA ICD code in OSA cases, and within the year of the last VA encounter in OSA controls.

Genome-wide association study in MVP and replication testing in FinnGen

Information on genotyping, imputation, and relatedness inference is provided in [Supplemental Note S1](#). Thorough details were provided elsewhere.^{28,29} We performed

association analyses using PLINK v2.00a3LM,³⁰ separately by HARE group and biological sex (confirmed by sex chromosomes), followed by meta-analysis using GWAMA v2.2.2³¹ with genomic control. As secondary findings, we also report results from analyses by HARE and by sex groups, and a multi-population meta-analysis stratified by obesity, and use them to study heterogeneity in associations across strata. Clumping of genome-wide significant SNPs to identify independent associations and annotation of top results are described in [Supplemental Note S2](#).

To test the detected associations for replication, we looked up top variants (after clumping) in publicly available GWAS of Finnish White individuals from FinnGen version 7 (N = 307,927: 27,207 cases and 280,720 controls). Summary statistics were downloaded from the FinnGen portal. Because the FinnGen OSA GWAS was not adjusted for BMI, we also downloaded summary statistics from a FinnGen GWAS of obesity. Then, we performed an approximated conditional analysis using the mt-COJO procedure implemented in GCTA v1.93.2.³² This resulted in GWAS summary statistics for OSA adjusted for obesity status. Because mt-COJO requires an LD reference panel, we used the 1000 Genomes Europeans (CEU) as well as the pre-computed LD scores based on the CEU population. An association was deemed replicated if its one-sided p-value, with direction determined by the discovery direction of association,³³ was < 0.05 . One-sided replication analysis is useful because it incorporates information from the discovery study in the replication inference: an association cannot be determined as “replicated” if it seems protective in one dataset and risk increasing in another dataset. We also report false discovery rate (FDR) adjusted replication p-values, computed using the Benjamini-Hochberg procedure³⁴ over all primary multi-population BMI-adj and BMI-unadj associations. Finally, we performed association look-ups in a GWAS of snoring from the UK Biobank.¹⁶

Follow-up to MVP analysis

As described in [Fig. 1](#), we performed several follow-up analyses based on MVP results. Detailed descriptions are provided in [Supplemental Note S3](#). Briefly, we constructed genetic risk scores (GRSs) based on BMI-adj and BMI-unadj GWAS in the MGB Biobank comparing estimated associations with OSA (the MGB Biobank methods are provided in [Supplemental Note S4](#)); evaluated heterogeneity of heritability and of estimated associations across HARE groups and biological sex (described below); estimated genetic correlations of OSA with other phenotypes (described in [Supplemental Note S5](#)) and assessed the impact of BMI on genetic correlations; evaluated pleiotropy of top OSA SNPs, and performed gene-based analyses. Access to individual-level MGB Biobank data is available to MGB investigators.

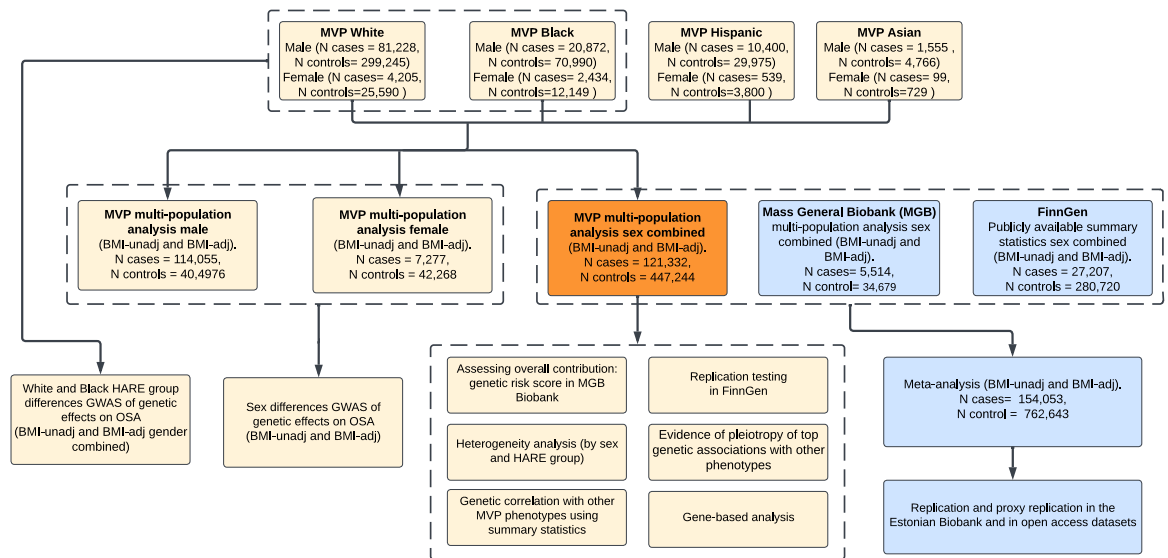


Fig. 1: Flowchart of the performed genetic analyses of OSA. The primary analysis is the multi-population OSA GWAS in MVP (orange colour box). This multi-population GWAS was a meta-analysis of sex- and HARE-group stratified analyses (top row of boxes). Various analyses were performed using the primary results, shown in the arrow coming down of the orange box. MVP analyses within sex strata and within Black and White HARE groups were used for assessment of potential group differences (two left most boxes in the bottom row). The MVP OSA GWAS summary statistics were also meta-analysed with GWAS summary statistics from MGB Biobank and FinnGen, followed by replication and proxy-replication look up in the Estonian Biobank and in open-access datasets (blue boxes). OSA: obstructive sleep apnoea. MVP: million veteran program. BMI: body mass index. BMI-adj: BMI-adjusted. BMI-unadj: BMI-unadjusted. HARE: harmonized race/ethnicity and genetic ancestry. MGB: Mass General Brigham.

Analysis of the heterogeneity of genetic associations with OSA across HARE group and sex

We analysed the evidence for genetic heterogeneity of OSA across HARE group and sex, in accordance with the known phenotypic heterogeneity. First, we estimated the heritability of OSA across HARE and biological sex groups. Heritability estimation is described below. Next, we tested differences in genetic associations between males and females, and between Black and White HARE groups (sex combined). Because all individuals were unrelated, we used as a test statistic for the j th genetic variant:

$$t_{s1-s2,j} = \frac{\beta_{s1,j} - \beta_{s2,j}}{\sqrt{SE_{s1,j}^2 + SE_{s2,j}^2}}$$

where $\beta_{s1,j}$, $\beta_{s2,j}$ are the estimated log-OR of the rare allele of the variants in strata $s1, s2$ and $SE_{s1,j}$, $SE_{s2,j}$ are the corresponding estimated SEs. We annotated genome-wide significant sex-different and HARE-different SNPs by manual extraction of associations with p -value < 0.001 from the Type 2 Diabetes Knowledge Portal.

We also evaluated the evidence of associations generalizing across discovery strata. For each sex, HARE, and obesity stratum, as well as the multi-population analysis, we identified SNP genome-wide

significant associations, and evaluated the evidence for their associations in other strata. Thus, we tabulated how many of the SNPs had p -value $< 5 \times 10^{-8}$ or < 0.05 in other strata. In addition, to account for differences in power due to sample sizes and allele frequencies, we followed Huang et al., 2022³⁵ and computed the power for identifying associations in the given stratum at the p -value < 0.05 level, and, for each SNP category defined by discovery GWAS (e.g., multi-population, male, Hispanic, etc.), we summed the estimated power to produce the expected number of generalized associations.

Meta-analysis with FinnGen and MGB Biobank

We meta-analysed the GWAS of multi-population individuals from MVP, the FinnGen OSA GWAS, and GWAS of (mostly White) unrelated individuals from the Mass General Brigham (MGB) Biobank (N BMI-unadj = 40,193, N BMI-adj = 35,960) using GWAMA v2.2.2.³¹ We performed clumping (1000 Kbp, $R^2 = 0.1$) to define the top hits using MVP multi-population genotypes as a reference panel. Next, we performed replication and proxy-replication look-up of the top hits using the following datasets: GWAS of sleep apnoea (BMI-adj and BMI-unadj) in the Estonian Biobank (described in Supplemental Note S6); GWAS of OSA from the PAN-UKBB GWAS summary statistics database; GWAS of OSA in a Han Chinese population¹¹; GWAS of snoring (BMI-adj and BMI-unadj) from the

UKBB¹⁶; GWAS of apnoea-hypopnea index,⁹ minimum and average oxyhaemoglobin saturation during sleep,²⁵ where the latter three are from meta-analyses of multiple, largely overlapping cohorts.

Ethics statement

MVP received ethical/study protocol approval from the VA Central Institutional Review Board, and written informed consent was obtained for all participants.

Statistics

Sample size was selected based on data availability. There was no randomization nor blinding, as this is an observational study. Statistical methods and software are described throughout the methods text and the [Supplementary Materials](#). In brief, GWAS in MVP tested SNP associations with OSA using the one degree of freedom Wald test as implemented in the PLINK software. Meta-analyses were performed on the log-OR scale using the inverse-variance fixed-effect meta-analysis, followed by the Wald test, as implemented by GWAMA. Test statistics used for group differences (sex differences in genetic associations and differences between associations in White and Black HARE groups) are described in the Methods section, subsection “Analysis of the heterogeneity of genetic associations with OSA across HARE group and sex”. P-value computation relied on asymptotic normality, and were implemented in the R software. We used the Wald test to test for the associations of OSA GRSs with OSA in the MGB Biobank, as implemented in the R ‘glm’ function.

Role of funders

The funding source had no involvement with the design, analysis, or summary of the work presented in this manuscript.

Results

Data from $N = 568,576$ contributed to the primary MVP dataset, with an overall OSA prevalence of 21.3%. Participants were 64 years old on average (standard deviation 15 years). 91.3% of the participants are male. [Fig. 2](#) visualizes the prevalence of OSA in the MVP dataset, and the distribution of BMI, stratified by HARE and biological sex groups. OSA is most prevalent in the Hispanic male group (25.6%), and least common in the Asian female group (11.9%). [Supplemental Table S1](#) characterizes the MVP study population by HARE group and combined. We performed genetic analysis of OSA according to the flowchart depicted in [Fig. 1](#).

Primary GWAS and replication

[Fig. 3](#) panel a provides a Miami plot visualizing the findings and annotating the nearest gene for each association region. QQ-plots are provided in [Supplementary Fig. S1](#) and regional association plots

are provided in [Supplementary Fig. S2](#) (BMI-unadj) and [S3](#) (BMI-adj). In the primary GWAS, there were 18 genome-wide significant association regions in the BMI-unadj analysis, and 6 in the BMI-adj analysis ([Table 1](#)). Of these, from the BMI-unadj analysis, 2 associations were not available in FinnGen (nor other genome-wide significant SNPs from the same regions), and 13 associations replicated (one-sided p -value < 0.05). Notably, all associations had the same direction of associations in MVP and in FinnGen. Of the 6 associations identified in BMI-adj analysis, 4 replicated in FinnGen and only one association had opposite direction of association between MVP and FinnGen ([Table 1](#)). Information about replicated associations and what was previously known about replicated SNPs is provided in [Supplemental Note S7](#). [Supplemental Table S2](#) reports previously reported associations of SNPs underlying genome-wide significant associations of OSA from [Table 1](#) with sleep traits, and summarizes the evidence of replication of associations. [Supplemental Table S3](#) reports the evidence of association of the OSA SNPs with snoring.¹⁶

We further assessed replication and generalization of estimated associations by constructing genetic risk scores (GRSs) for OSA in the MGB Biobank based on the associations in [Table 1](#). MGB Biobank participant characteristics are provided in [Supplementary Table S4](#). Complete association analysis results are provided in [Supplementary Table S5](#) and details are provided in [Supplemental Note S8](#). [Fig. 4](#) demonstrates that while GRS based on both BMI-unadj and BMI-adj GWAS associated with OSA in the MGB biobank, once adjusting the analysis to BMI, only the GRS based on BMI-adj GWAS is associated with OSA.

Heterogeneity of genetic associations with OSA

[Supplementary Fig. S4](#) demonstrates that estimated heritability tends to be lower in the multi-population analysis compared to each of the HARE group-specific estimates, and similarly lower in sex-combined analysis compared to sex-specific analyses. [Supplementary Fig. S5](#) provides the Manhattan and qq-plots from sex-differences GWAS, demonstrating widespread significant sex differences in genetic associations with OSA. [Fig. 5](#) visualizes the estimated associations (odds ratios) of the minor alleles in males and in females across categories of SNPs defined by their p -value in test of sex differences, based on BMI-adj analysis. [Supplementary Fig. S6](#) provides similar results for BMI-unadj analysis. For the variants with most significant associations, all estimated effects in females are protective. We manually annotated the genome-wide significant sex-different SNPs (after clumping, 70 SNPs from BMI-unadj and 68 SNPs from BMI-adj analysis, partly overlapping) using the Type 2 Diabetes Knowledge Portal, reporting associations with p -values < 0.001 with any trait ([Supplementary Table S6](#)). [Supplementary Fig. S7](#)

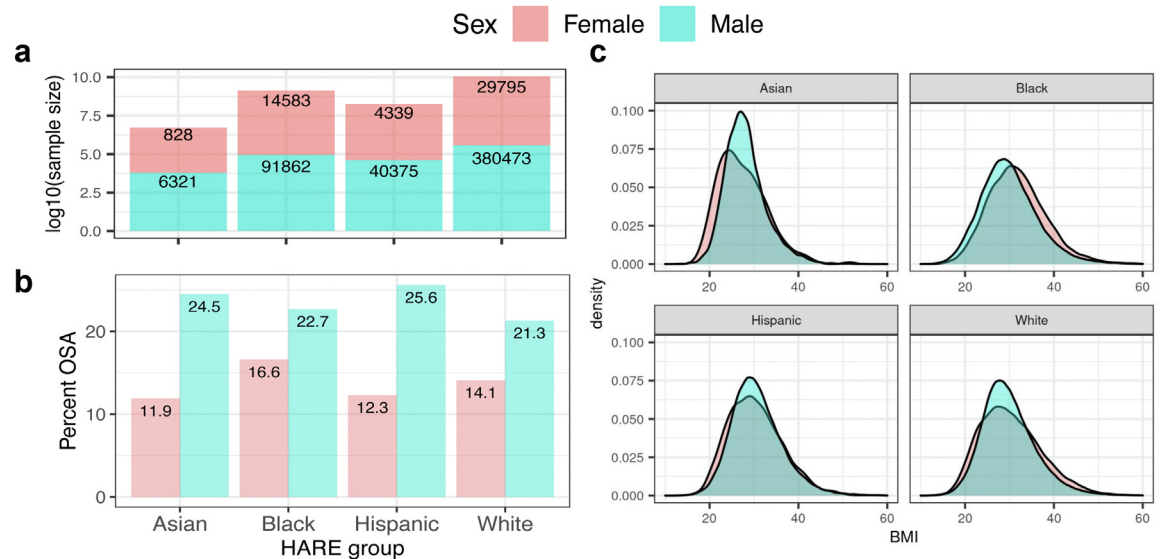


Fig. 2: Proportions of OSA and BMI distribution across MVP HARE groups. The figure provides three panels. The top left panel visualizes the number of MVP study participants by HARE group and by biological sex. The bottom left panel visualizes the percentages of MVP participants who have OSA according to the EHR-based phenotype, stratified by HARE group and by sex. The right panel provides density plots visualizing the distributions of participants' BMI, stratified by HARE and sex. OSA: obstructive sleep apnoea. MVP: million veteran program. HARE: harmonized race/ethnicity and genetic ancestry. BMI: body mass index. EHR: electronic health records.

visualizes the number of previously reported SNP associations at this significance level for OSA sex-different SNPs by trait category, based on BMI-adj and BMI-unadj analyses.

Supplementary Tables S7 and S8 report results from BMI-unadj and from BMI-adj analyses of sex-combined and sex-specific gene-based analyses of the multi-population OSA GWAS. In males, the second strongest association in BMI-unadj analysis was of the *ETV5* (ETS Variant Transcription Factor 5) gene, which had no evidence of association in females (MAGMA p-value = 0.23). This gene is involved in cellular response to oxidative stress³⁶ and is mostly expressed in the pituitary gland according to the gene-tissue expression (GTEx) portal.³⁷ The top association in females and only association with FDR p-value < 0.05 was with the *SPIDR*, Scaffold Protein Involved in DNA Repair, gene (FDR = 0.04; MAGMA p-value = 2.3×10^{-6}) and was not associated with OSA in males (MAGMA p-value = 0.08). In GTEx, *SPIDR* is mostly expressed in ovary tissue and in the cerebellar hemisphere.

We also compared SNP associations between the White and Black HARE groups, as the two largest groups. Manhattan and qq-plots are reported in Supplementary Fig. S8, and top results are provided in Supplementary Table S9. After clumping of SNPs, there are 8 and 12 genome-wide significant associations in BMI-adj and BMI-unadj analysis, respectively. Seven of the SNPs were previously associated (p-value < 0.001) with a range of phenotypes, including glycaemic and anthropometric traits, and snoring.

Finally, Supplementary Tables S10 and S11 report genome-wide significant SNP associations based on each of the main and secondary analyses, and further reports their associations across all strata-specific analyses. Supplementary Fig. S9 demonstrates that the number of generalizations across groups at the 0.05 p-value level always exceeded the expected number of generalizations computed using power analysis. More detailed results from assessment of heterogeneity of OSA genetic associations are provided in Supplemental Note S9.

Genetic correlations of OSA with cardiometabolic phenotypes and evidence of pleiotropy with and without accounting for BMI

We estimated the genetic correlation between OSA and a range of phenotypes. The results are visualized in Fig. 6. The estimated genetic correlations between OSA and blood pressure, lipids, and diabetes reduced substantially once accounting for BMI and often lost statistical significance. There were multiple statistically significant genetic correlations between OSA and sleep traits, and these often substantially changed between BMI-adj and BMI-unadj analysis. We also considered the evidence of pleiotropy in genome-wide significant OSA SNPs (Supplementary Tables S12 and S13). At the genome-wide significance level, in BMI-unadj analysis many OSA-SNP associations were identified with other traits, including insomnia/sleepiness, blood pressure, coronary artery disease, lipids, glycaemic traits, as well as alcohol use and smoking. In BMI-adj analyses, none

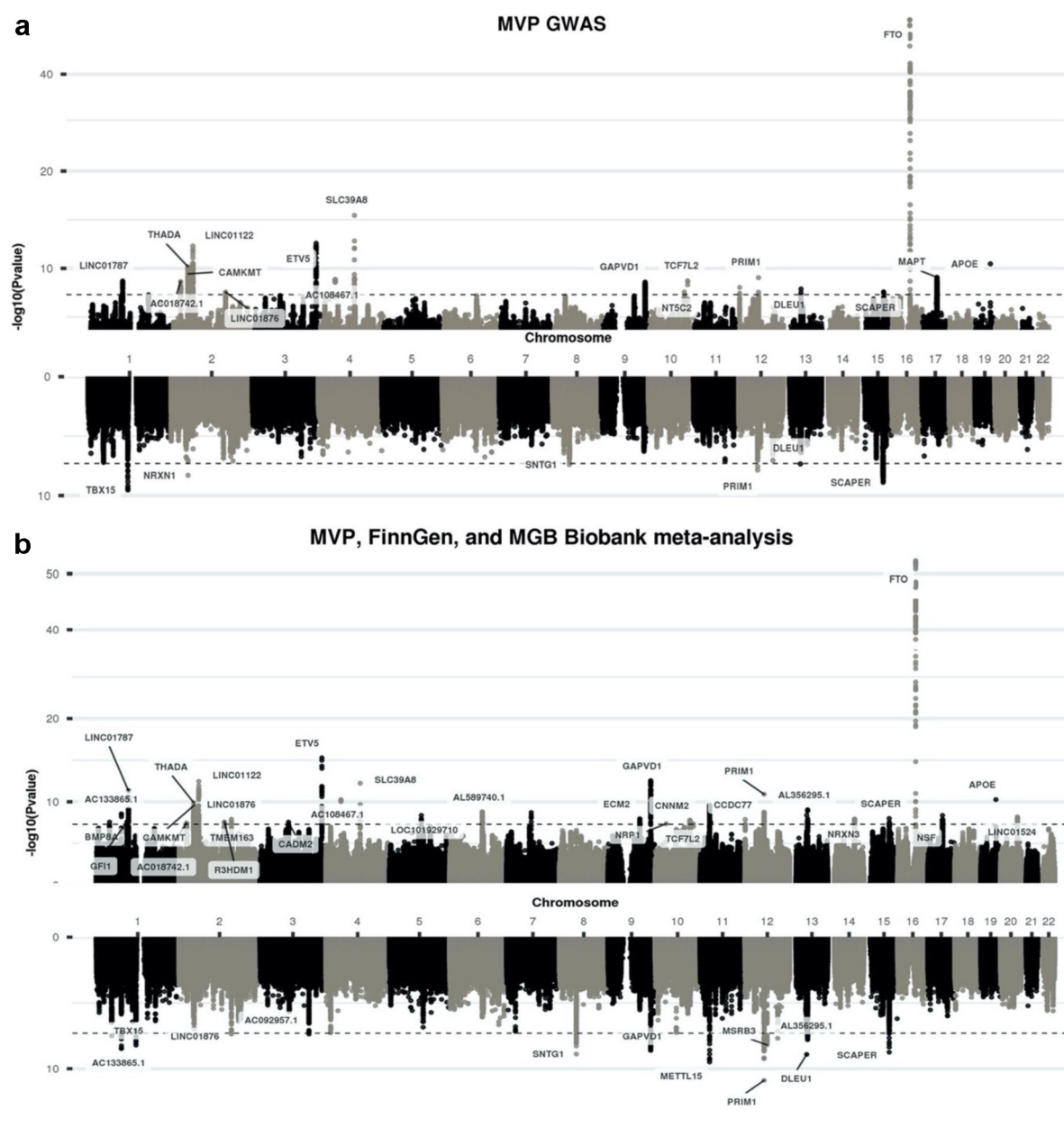


Fig. 3: Miami plots of the main MVP analyses and of the MVP, FinnGen, and MGB Biobank meta-analyses. Panel (a) provides a Miami plot from the primary MVP GWAS and panel (b) provides a Miami plot from the meta-analysis of MVP, FinnGen, and MGB Biobank. In each Miami plot, the top part provides results from BMI-unadj analysis, and the bottom from BMI-adj analysis. Each point corresponds to a SNP, with position on the x-axis determined by chromosomal location (genome build hg37) and position on the y-axis corresponding to the negative of the base-10 log of the association p-value of that SNP in association with OSA and in the respective analysis. Thus, sets of points that appear to be vertically aligned correspond to sets of SNPs that are likely in linkage-disequilibrium with each other within the same genomic region. These vertically aligned points are annotated by the nearest gene. P-values were computed based on the 1-degree of freedom Wald test. MVP: million veteran program. GWAS: genome-wide association study. MGB: Mass General Brigham. SNP: single nucleotide polymorphism. BMI: body mass index. BMI-unadj: BMI-unadjusted. BMI-adj: BMI-adjusted.

of the OSA SNPs is associated with the considered outcomes at the 5×10^{-8} level. Additional information and interpretation of results is provided in [Supplemental Note S10](#).

Secondary GWAS: meta-analysis of MVP and other biobank-based GWAS

We meta-analysed the primary GWAS from MVP, the publicly available FinnGen Freeze 7 GWAS summary

CHR	rsID	Nearest gene	Position	A1	A2	MVP discovery			FinnGen replication				
						EA	OR	P-value	EA	OR	P-value	One-sided P-value	FDR BH
BMI-unadjusted													
1	rs17367240	LINC01787	96960057	A	G	0.38	0.97	2.07E-09	0.29	0.97	1.33E-03	6.65E-04	1.77E-03
2	rs58857776	LINC01122	58890684	T	C	0.41	0.96	5.06E-13	0.51	0.97	2.79E-03	1.40E-03	2.48E-03
2	rs77881454	THADA	43757293	T	A	0.11	1.05	6.72E-11	0.07	1.06	1.91E-03	9.53E-04	1.98E-03
2	rs540606	CAMKMT	45138507	A	G	0.39	1.03	3.71E-10	0.36	1.02	4.12E-02	2.06E-02	3.00E-02
2	rs11691026	AC018742.1	22131865	A	G	0.44	1.03	2.32E-09	0.51	1.01	3.09E-01	1.54E-01	1.65E-01
2	rs181355045	LINC01876	156998417	A	C	0.11	1.05	2.78E-08	0.12	1.04	1.98E-03	9.90E-04	1.98E-03
3	rs869400	ETV5	185826740	T	G	0.20	0.95	2.63E-13	0.16	0.95	1.34E-04	6.72E-05	2.15E-04
4	rs13107325	SLC39A8	103188709	T	C	0.07	1.09	3.58E-16	0.01	1.04	2.93E-01	1.46E-01	1.65E-01
4	rs10938397	AC108467.1	45182527	G	A	0.39	1.03	1.37E-09	0.47	1.04	5.16E-05	2.58E-05	1.03E-04
9	rs5015933	GAPVD1	128137418	C	T	0.51	0.97	2.61E-09	0.55	0.95	1.91E-08	9.55E-09	7.64E-08
10	rs34872471	TCF7L2	114754071	C	T	0.30	0.97	1.93E-09	0.20	0.98	5.86E-02	2.93E-02	3.61E-02
10	rs201575845	NT5C2	104861271	A	AT	0.10	1.05	2.72E-08	NA	NA	NA	NA	NA
12	rs2277339	PRIM1	57146069	G	T	0.12	0.95	9.05E-10	0.13	0.93	3.44E-07	1.72E-07	9.17E-07
13	rs201770	DLEU1	51047531	G	A	0.22	0.97	1.32E-08	0.30	1.00	9.55E-01	5.23E-01	5.23E-01
15	rs283789	SCAPER	76777577	G	A	0.36	0.97	2.61E-08	0.33	0.98	5.31E-02	2.65E-02	3.54E-02
16	rs1558902	FTO	53803574	A	T	0.33	1.08	3.80E-47	0.42	1.10	1.54E-23	7.70E-24	1.23E-22
17	rs58879558	MAPT	44095467	C	T	0.20	1.04	8.22E-10	NA	NA	NA	NA	NA
19	rs429358	APOE	45411941	C	T	0.15	0.95	3.40E-11	0.18	0.97	1.75E-02	8.73E-03	1.40E-02
BMI-adjusted													
1	rs79950770	TBX15	119539797	T	C	0.09	1.06	2.93E-10	0.16	1.03	6.15E-03	3.08E-03	4.61E-03
2	rs72618639	NRXN1	50795733	G	C	0.02	0.88	5.02E-09	0.03	1.01	6.83E-01	6.58E-01	6.58E-01
8	rs34179442	SNTG1	50900444	A	G	0.47	0.97	3.74E-08	0.68	0.97	4.58E-03	2.29E-03	4.58E-03
12	rs2277339	PRIM1	57146069	G	T	0.12	0.96	1.38E-08	0.13	0.93	2.29E-08	1.14E-08	6.87E-08
13	rs592333	DLEU1	51340315	G	A	0.46	1.03	4.68E-08	0.52	1.04	4.48E-05	2.24E-05	6.72E-05
15	rs4886823	SCAPER	76965402	G	A	0.27	0.96	1.30E-09	0.24	0.99	3.06E-01	1.53E-01	1.84E-01

CHR: chromosome; Position: genomic coordinate in genome build hg37; A1: effect allele; A2: non-effect allele; OR: estimated odds ratio per one effect allele; P-value: two-sided p-value computed based on the Wald test using a reference normal distribution. MVP sample size: N = 568,576 in BMI-unadj analysis; N = 559,070 in BMI-adj analysis. FinnGen sample size: 27,207 cases, 280,720 controls. One-sided p-values were computed based on the direction of association observed in the discovery MVP study, so that no association may be replicated if the direction of association does not match across discovery and replication. FDR BH refers to False Discovery Rate adjusted p-values computed using the Benjamini-Hochberg method, applied over the one-sided p-values.

Table 1: Top variants associated with OSA in multi-ethnic analysis (sex combined) in BMI-adj and -unadj analyses.

statistics, and a GWAS of OSA from the MGB Biobank. [Fig. 3](#) panel b provides the Miami plot, and [Supplementary Table S14](#) provides the top (clumped, genome-wide significant) associations. There were 31 and 12 genome-wide significant associations with OSA in BMI-unadj and BMI-adj analyses, respectively. [Supplementary Tables S15 and S16](#) provide replication and proxy-replication look ups of the genome-wide significant SNPs in GWAS of sleep apnoea, snoring, and measures of sleep disordered breathing obtained using at-home sleep studies. [Supplementary Table S17](#) summarizes the overlap between this GWAS, the MVP-only GWAS, the previously published FinnGen GWAS,⁶ and the OSA GWAS using multi-trait analysis with snoring,¹⁷ and the evidence of replication and proxy-replication. This analysis detected 15 associations regions that were not previously reported in the other GWAS. Of these, 5 regions had evidence of association in the Pan UKBB sleep apnoea GWAS, and 5 additional regions had evidence of association with snoring. While the GWAS of sleep apnoea in the Estonian biobank did

not provide any strong replication evidence, it supported the replication evidence in 6 of the 10 regions, by having the same direction of association as the discovery GWAS and nominally significant two-sided p-values (<0.05).

Discussion

We performed GWAS of OSA in a large dataset of N = 568,576 MVP participants, and further performed a meta-analysis of MVP, FinnGen, and the MGB Biobank for a total of 916,696 individuals. We see evidence that many OSA genetic loci are associated with other sleep phenotypes, affirming the contribution of OSA to impaired sleep. We also identified and replicated or proxy-replicated (using other sleep phenotypes) multiple loci that were not yet reported for sleep phenotypes. We show that a GRS based on BMI-adj OSA GWAS is associated with OSA even when adjusting for BMI in an independent dataset, confirming a role of genetics in OSA aetiology independent of BMI. Finally, we see pervasive sex differences in genetic associations with OSA.

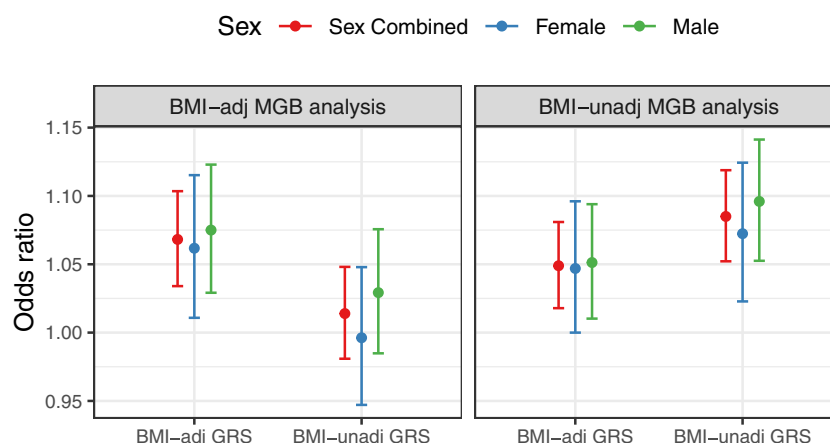


Fig. 4: Association of OSA GRSs with OSA in the MGB Biobank. The figure provides estimated ORs and accompanying 95% confidence intervals of OSA per 1 standard SD increase in OSA GRSs constructed based on BMI-adjusted GWAS ("BMI-adj GRS") and based on BMI-unadjusted GWAS ("BMI-unadj GRS"). The GWAS used to guide the construction of the GRSs are the primary, multi-population MVP GWAS, and were constructed as weighted sums of all SNPs reported in Table 1. Thus, SNP alleles were weighted by the estimated log-OR in MVP (Table 1). The GRSs were constructed, and association analyses were performed in the MGB biobank. Association analyses in MGB biobank were performed using logistic regression while adjusting to BMI (BMI-adj MGB analysis; left) and without adjusting to BMI (BMI-unadj MGB analysis; right). All MGB analyses were adjusted to age, sex, genetic batch, and the first 10 PCs of genetic data. OR: odds ratio. OSA: obstructive sleep apnoea. SD: standard deviation. GRS: genetic risk score. BMI: body mass index. BMI-adj: BMI-adjusted. GWAS: genome-wide association study. MVP: million veteran program. SNP: single nucleotide polymorphism. MGB: Mass General Brigham. PC: principal component.

Almost all OSA variant associations have some level of associations in a GWAS of snoring from the UK Biobank.¹⁶ There were also a few SNP associations that were not associated with snoring in the UKB GWAS. For example, the A allele of rs1373285 was associated with increased OSA risk in the MVP, FinnGen, and MGB Biobank meta-analysis in BMI-adj analysis and the association was replicated in the Pan-UKBB GWAS of sleep apnoea. The A allele of this SNP was previously associated with higher odds or frequent insomnia symptoms in a UKB GWAS.³⁸ This suggests that OSA may lead to insomnia, likely due to poor sleep quality, in some individuals who may not snore. Another SNP, rs4953923, was associated with OSA in BMI-unadj analysis and did not proxy-replicate for snoring. It was weakly associated with OSA in the Pan-UKBB GWAS (p -value = 0.04). Its strongest previously reported association was with chronotype: T allele was associated with lower likelihood of being a morning person.³⁹ Rs13107325 was discovered and replicated in the Pan-UKBB OSA GWAS, with its T allele increasing risk of OSA. This SNP as well was not associated with snoring (p -value = 0.36) but it was associated with both short sleep duration (2.5×10^{-13}) and with a continuous measure of sleep duration,⁴⁰ with the T allele associated with shorter sleep. The protein encoded by this gene is found in the mitochondria, and has a role in the cellular import of zinc at the onset of inflammation.⁴¹ More work is needed to identify commonalities and differences in genetic underpinnings of sleep traits.

Gender and sex differences are well known for OSA. Previous work reported that the association of OSA with body fat distribution measures differ by sex,⁴² and gender differences in OSA phenotypes, such as respiratory event length and apnoea-hypopnea index during non-REM sleep.^{43,44} This may be on top of well-established gender and sex differences in anthropometric measures,⁴⁵ and in the genetics underlying such measures, e.g., BMI, waist-to-hip ratio, neck circumferences, and others.^{46,47} Mechanisms leading to gender and sex differences in genetic associations may include amplification of gene expression due to hormone levels, contributions of the X-chromosome, differential genetic liability distributions, and gene-environment interaction, where exposures that tend to have differential distribution across genders modify genetic effects.^{48,49} In this work we see genetic differences across sex groups throughout the autosomes. Annotation of the genome-wide significant sex-different OSA variants based on previously-published associations did not reveal specific patterns, though we saw many glycaemic trait and cardiovascular-related associations. This observation may be related to reported sex differences in diabetes sensitivity.⁵⁰ Notably, in previous work we saw that glycaemic traits have a causal effect on OSA, as suggested by Mendelian randomization analysis.⁵¹ Thus, gender and sex differences in glycaemic traits may be related to the observed sex differences in OSA genetics. Future work should further dissect gender and sex differences in OSA and its determinants, and study the role of sex chromosomes in sex differences in OSA.

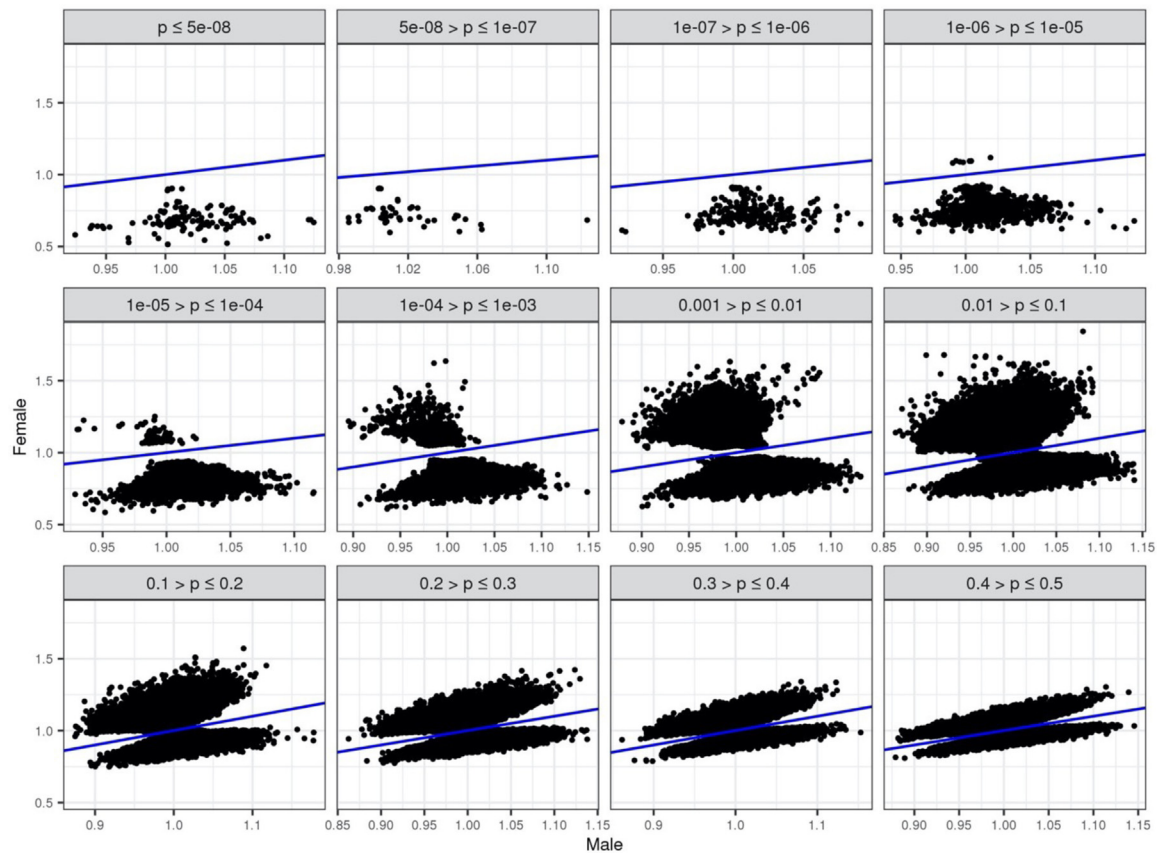


Fig. 5: Protective genetic associations with OSA in females in SNPs with evident sex interactions. The figure compares the estimated OR of SNPs in association with OSA in male and in female participants from MVP. Each square corresponds to a set of SNPs defined by their p-value (p) in test of difference in their effect sizes between biological sex groups. SNPs with sex-difference effect p-values > 0.5 are not shown. Depicted results are from BMI-adj analysis. The blue line is the line of identity. The test statistics used to compute p-values are provided in the main manuscript. P-values were two-sided p-values computed assuming that the test statistics has standard normal distribution. OR: odds ratio. SNP: single nucleotide polymorphism. OSA: obstructive sleep apnoea. MVP: million veteran program. P: p-value. BMI: body mass index. BMI-adj: BMI-adjusted.

We studied the genetic evidence for the heterogeneity of OSA across HARE groups defined by race/ethnicity and genetics, as well as by obesity. First, heritability estimates within strata tend to be higher compared to estimates computed over groups that combine multiple strata (multi-population, as well as combined sex groups). A similar phenomenon was previously reported³² in a study of sex differences: the authors reported that estimated heritability in sex-combined analysis tended to be lower compared to estimated heritability within sex groups, and this pattern became more pronounced when the genetic correlations between male and female genetic effects were lower. Thus, the heritability analysis supports the existence of some differences in OSA genetic architecture between groups. Yet, in an analysis weighing the evidence for generalization of genome-wide significant associations across strata, when considering the power to detect associations, we could not observe

any difference between groups. In contrast, rates of generalizations at the p-value < 0.05 were sometimes higher than expected according to the power analysis. Nevertheless, this analysis is limited because the majority of MVP participants (67%) are White males, so that power to discover associations that are specific to other groups is lower. Our analysis of differences between White and Black HARE groups OSA associations did identify a handful of associations, though to a much lesser degree compared with sex-different OSA SNPs. Critically, these differences can be due to different exposures modifying genetic effects, i.e., gene by environment interactions. In fact, two of the SNP associations that had HARE differences were within OSA-associated genes: *FTO* and *PRIM1*, suggesting that there may be differences in effect size of known genes. This could also be due to gene-gene interactions (and not only gene-environment interactions).

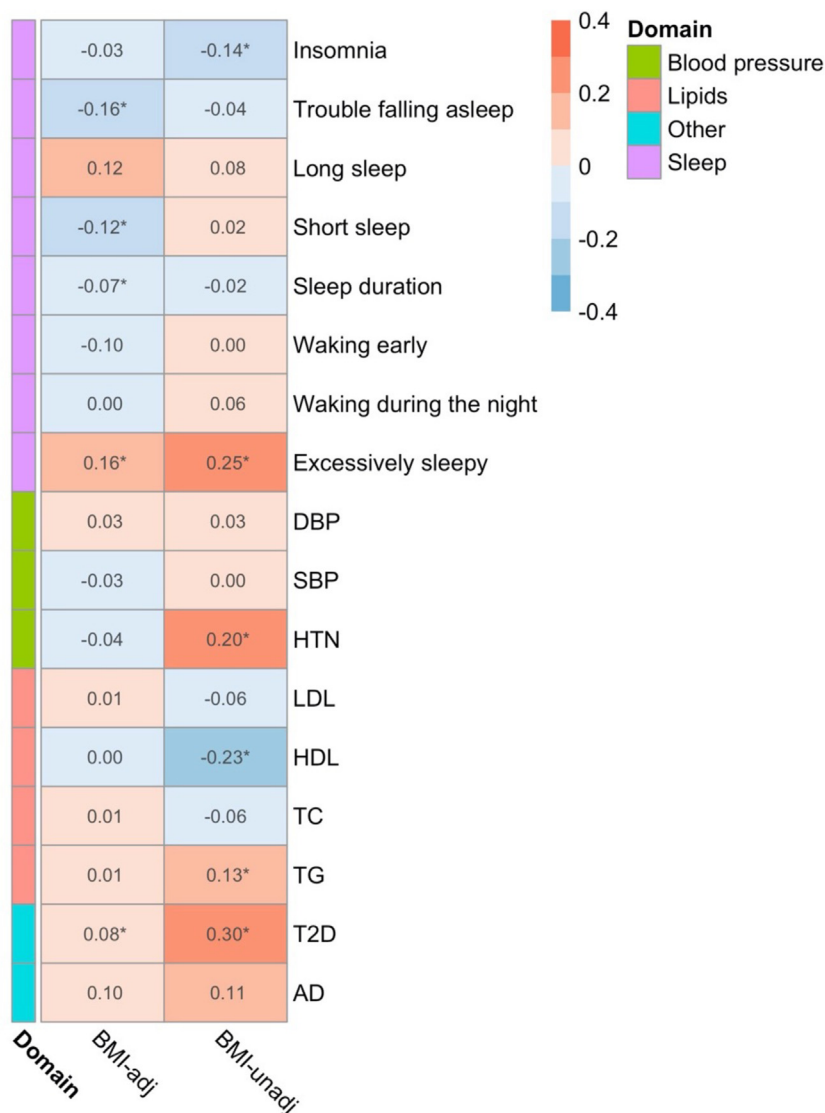


Fig. 6: Genetic correlations of OSA with cardiometabolic traits are driven by BMI. Visualization of the estimated genetic correlations between OSA and other phenotypes, in BMI-adj and BMI-unadj analyses. For each pair of phenotypes, genetic correlation was computed using LDSC based on summary statistics from GWAS, restricted to HapMap SNPs. Phenotypes are described in [Supplemental Note S5](#). * indicates LDSC p-value < 0.05. OSA: obstructive sleep apnoea. BMI: body mass index. BMI-adj: BMI-adjusted. BMI-unadj: BMI-unadjusted. LDSC: Linkage disequilibrium score regression. GWAS: genome-wide association study. SNP: single nucleotide polymorphism. DBP: diastolic blood pressure; SBP: systolic blood pressure; HTN: hypertension; LDL: low-density lipoprotein cholesterol; HDL: high-density lipoprotein cholesterol; TC: total cholesterol; TG: triglycerides; T2D: type 2 diabetes; AD: Alzheimer's disease. HapMap SNPs: tag SNPs identified by the haplotype mapping (HapMap) project as uniquely able to identify haplotype blocks across multi-ancestry populations.

Strengths of this study include large sample size, high prevalence of OSA that is close to the population level rather than the lower level often observed in biobanks, where lower levels are due to underdiagnosis and thus cases are often misclassified as controls. Also, we performed BMI-adj analysis, improving upon the largest previously-published GWAS of OSA from FinnGen⁶ that did not provide genome-wide results from BMI-adj analysis. We further studied the genetic correlation

between OSA and cardiometabolic and other phenotypes with and without accounting for BMI. Another strength is the investigation of sex differences. Study limitations include the population not being representative: participants are U.S. Veterans and predominately males. This has implications for the study of sex differences: as, further, OSA prevalence is lower in females compared to males, power for discovery of genetic associations in females is substantially lower compared to

males and results may not generalize as well for females. It is reassuring that SNP-level replication study was performed in the population-based FinnGen biobank study, where men are not over-represented, and that GRS based on genome-wide significant SNPs identified in MVP was well associated with OSA in both male and female participants from the MGB biobank. Yet, it is possible that the few non-replications of individual-SNP associations are due to sex imbalance between the MVP discovery and FinnGen replication studies. Sex imbalance also has implication of genome-wide analysis of sex differences in genetic effects. However, we note that this analysis appropriately accounted for differences in sample sizes across sex strata using sex-specific standard errors, which tend to be higher in the stratum of women due to the lower sample size. Another limitation is that phenotypes are based on medical records, and as such are prone to various selection biases. For example, some individuals may be treated in other healthcare systems so that covariate and treatment information may not be complete. Individuals are possibly underdiagnosed for OSA as well as other comorbidities, due to self- or doctor- selection for diagnosis and treatment. However, importantly, the VA healthcare system provides free diagnosis and treatment for OSA if needed, therefore the likely proportion of patients with undiagnosed OSA among controls is expected to be lower compared to other biobanks, as is reflected in a prevalence of diagnosed OSA that is closer to the expected population prevalence of this disorder than in previous biobank studies. Another limitation is the use of a dichotomized OSA trait, reflecting the limited availability of more granular severity measures in biobank studies. This is likely to reduce power to detect true associations but should not otherwise bias the analysis.

This study adds to an increasing literature of OSA genetics, including studies using medical records or self-reported phenotypes,^{6,17} that are likely biased and lose power by misclassification of cases and controls⁵³ but have hundreds of thousands of individuals; studies utilizing pleiotropy of OSA with other phenotypes to increase analysis power for discovery of genetic effects^{17,54}; population-based studies in smaller sample sizes of thousands or a few tens of thousands of individuals with carefully collected severity measures of OSA^{9–11,55}; and smaller clinic-based studies typically measuring OSA phenotypes in hundreds of individuals, typically targeting specific candidate genes or variants.^{56–59} Future work should evaluate and synthesize information across OSA genetic studies to identify replication across studies, obtain population-level effect size estimates, and assess selection biases potentially limiting reported findings.

In summary, we report a large GWAS of OSA in MVP, assessing in detail the genetic heterogeneity of

OSA across sex and HARE groups, and its relation to other phenotypes, and a second GWAS meta-analysis of MVP, FinnGen, and the MGB Biobank, leveraging large available sample sizes for increased discovery power. We observed pervasive sex differences in genetic associations, and these sex-different associations were generally protective against OSA in women, providing a possible genetic window to understanding the greater prevalence of OSA in men than in women. Future work should assess and validate the functionality of newly discovered genome-wide significant OSA loci, estimate the associations of discovered loci and genes with OSA severity measures and with continuous measures characterizing OSA such as the apnoea-hypopnea index, and build upon these findings to develop strategies to identify undiagnosed individuals who likely have OSA and may benefit from treatment, potentially in a sex-specific manner.

Contributors

Study conceptualisation: T Sofer, DJ Gottlieb. Data curation: M Murray, Y-L Ho, J Huffman, K Cho, PWF Wilson, Estonian Biobank Research Team. Formal analysis: N Kurniansyah, M Murray, Erik Abner. Funding acquisition: K Cho, PWF Wilson, Tõnu Esko. Supervision: T Sofer. Visualisation: N Kurniansyah. Writing – original draft: T Sofer, N Kurniansyah. Writing – review & editing: M Murray, Y-L Ho, J Huffman, K Cho, PWF Wilson, DJ Gottlieb, Erik Abner, Tõnu Esko. T Sofer and N Kurniansyah have directly accessed and verified the underlying data reported in the manuscript. All authors had full access to all the data in the study and accept responsibility to submit for publication.

Data sharing statement

Full GWAS summary statistics are being deposited on dbGaP (<https://www.ncbi.nlm.nih.gov/gap/>) under the MVP accession (phs001672). FinnGen GWAS summary statistics (version 7) are available from the FinnGen portal (<https://www.finnngen.fi/en/public-release-finnngen-data-freeze-7-results-and-summary-statistics>). MGB Biobank data are available to MGB investigators. Summary statistics for sleep-related phenotypes were accessed through the Type 2 Diabetes Knowledge Portal (<https://t2d.hugeamp.org/>). Summary statistics of sleep apnoea GWAS from UKBB were accessed via the Broad Institute webpage (<https://pan.ukbb.broadinstitute.org/>).

Declaration of interests

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.ebiom.2023.104536>.

References

- Chan ASL, Phillips CL, Cistulli PA. Obstructive sleep apnoea—an update. *Intern Med J*. 2010;40(2):102–106.
- de Paula LKG, Alvim RO, Pedrosa RP, et al. Heritability of OSA in a rural population. *Chest*. 2016;149(1):92–97.
- Redline S, Tishler PV, Tosteson TD, et al. The familial aggregation of obstructive sleep apnea. *Am J Respir Crit Care Med*. 1995;151(3 Pt 1):682–687.
- Redline S, Tishler PV. The genetics of sleep apnea. *Sleep Med Rev*. 2000;4(6):583–602.
- Wang H, Cade BE, Sofer T, et al. Admixture mapping identifies novel loci for obstructive sleep apnea in Hispanic/Latino Americans. *Hum Mol Genet*. 2019;28(4):675–687.
- Strausz S, Ruotsalainen S, Ollila HM, et al. Genetic analysis of obstructive sleep apnoea discovers a strong association with cardiometabolic health. *Eur Respir J*. 2021;57(5):2003091.
- Farias Tempaku P, Leite Santoro M, Bittencourt L, D'Almeida V, Iole Belangero S, Tufik S. Genome-wide association study reveals two novel risk alleles for incident obstructive sleep apnea in the EPISONO cohort. *Sleep Med*. 2020;66:24–32.
- Baik I, Seo HS, Yoon D, Kim SH, Shin C. Associations of sleep apnea, NRG1 polymorphisms, alcohol consumption, and cerebral white matter hyperintensities: analysis with genome-wide association data. *Sleep*. 2015;38(7):1137–1143.
- Chen H, Cade BE, Gleason KJ, et al. Multiethnic meta-analysis identifies RAI1 as a possible obstructive sleep apnea-related quantitative trait locus in men. *Am J Respir Cell Mol Biol*. 2018;58(3):391–401.
- Cade BE, Chen H, Stimp AM, et al. Genetic associations with obstructive sleep apnea traits in Hispanic/Latino Americans. *Am J Respir Crit Care Med*. 2016;194(7):886–897.
- Xu H, Liu F, Li Z, et al. Genome-wide association study of obstructive sleep apnea and objective sleep-related traits identifies novel risk loci in Han Chinese individuals. *Am J Respir Crit Care Med*. 2022;206(12):1534–1545.
- Suarez E, Fang SC, Bliwise D, Yaggi HK, Araujo A. Disentangling racial/ethnic and socioeconomic differences in self-reported sleep measures: the Boston Area Community Health Survey. *Sleep Health*. 2015;1(2):90–97.
- Villaneuva ATC, Buchanan PR, Yee BJ, Grunstein RR. Ethnicity and obstructive sleep apnoea. *Sleep Med Rev*. 2005;9(6):419–436.
- Dudley KA, Patel SR. Disparities and genetic risk factors in obstructive sleep apnea. *Sleep Med*. 2016;18:96–102.
- Franklin KA, Lindberg E. Obstructive sleep apnea is a common disorder in the population—a review on the epidemiology of sleep apnea. *J Thorac Dis*. 2015;7(8):1311–1322.
- Campos AI, García-Marín LM, Byrne EM, Martin NG, Cuéllar-Partida G, Rentería ME. Insights into the aetiology of snoring from observational and genetic investigations in the UK Biobank. *Nat Commun*. 2020;11(1):817.
- Campos AI, Ingold N, Huang Y, et al. Discovery of genomic loci associated with sleep apnoea risk through multi-trait GWAS analysis with snoring. *Sleep*. 2022;46(3):zsac308.
- Kapur V, Strohl KP, Redline S, Iber C, O'Connor G, Nieto J. Underdiagnosis of sleep apnea syndrome in U.S. communities. *Sleep Breath*. 2002;6(2):49–54.
- Aurora RN, Quan SF. Quality measure for screening for adult obstructive sleep apnea by primary care physicians. *J Clin Sleep Med*. 2016;12(8):1185–1187.
- Kapur VK, Auckley DH, Chowdhuri S, et al. Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea: an American academy of sleep medicine clinical practice guideline. *J Clin Sleep Med*. 2017;13(3):479–504.
- Santilli M, Manciocchi E, D'Addazio G, et al. Prevalence of obstructive sleep apnea syndrome: a single-center retrospective study. *Int J Environ Res Public Health*. 2021;18(19):10277.
- Simpson L, Hillman DR, Cooper MN, et al. High prevalence of undiagnosed obstructive sleep apnoea in the general population and methods for screening for representative controls. *Sleep Breath*. 2013;17(3):967–973.
- Young T, Peppard PE, Gottlieb DJ. Epidemiology of obstructive sleep apnea: a population health perspective. *Am J Respir Crit Care Med*. 2002;165(9):1217–1239.
- Senaratna CV, Perret JL, Lodge CJ, et al. Prevalence of obstructive sleep apnea in the general population: a systematic review. *Sleep Med Rev*. 2017;34:70–81.
- Cade BE, Chen H, Stimp AM, et al. Associations of variants in the hexokinase 1 and interleukin 18 receptor regions with oxyhemoglobin saturation during sleep. *PLoS Genet*. 2019;15(4):e1007739.
- Folmer RL, Smith CJ, Boudreau EA, et al. Prevalence and management of sleep disorders in the veterans health administration. *Sleep Med Rev*. 2020;54:101358.
- Alexander M, Ray MA, Hébert JR, et al. The national veteran sleep disorder study: descriptive epidemiology and secular trends, 2000–2010. *Sleep*. 2016;39(7):1399–1410.
- Peloso GM, Tcheandjieu C, McGeary JE, et al. Genetic loci associated with COVID-19 positivity and hospitalization in white, black, and hispanic veterans of the VA million veteran program. *Front Genet*. 2021;12:777076.
- Hunter-Zinck H, Shi Y, Li M, et al. Genotyping array design and data quality control in the million veteran program. *Am J Hum Genet*. 2020;106(4):535–548.
- Chang CC, Chow CC, Tellier LC, Vattikuti S, Purcell SM, Lee JJ. Second-generation PLINK: rising to the challenge of larger and richer datasets. *GigaScience*. 2015;4:7.
- Mägi R, Morris AP. GWAMA: software for genome-wide association meta-analysis. *BMC Bioinformatics*. 2010;11:288.
- Yang J, Lee SH, Goddard ME, Visscher PM. GCTA: a tool for genome-wide complex trait analysis. *Am J Hum Genet*. 2011;88(1):76–82.
- Sofer T, Heller R, Bogomolov M, et al. A powerful statistical framework for generalization testing in GWAS, with application to the HCHS/SOL. *Genet Epidemiol*. 2017;41(3):251–258.
- Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J Roy Stat Soc B*. 1995;57(1):289–300.
- Huang QQ, Sallah N, Dunca D, et al. Transferability of genetic loci and polygenic scores for cardiometabolic traits in British Pakistani and Bangladeshi individuals. *Nat Commun*. 2022;13:4664.
- Monge M, Colas E, Doll A, et al. Proteomic approach to ETV5 during endometrial carcinoma invasion reveals a link to oxidative stress. *Carcinogenesis*. 2009;30(8):1288–1297.
- Carithers LJ, Moore HM. The Genotype-Tissue Expression (GTEx) project. *Biopreserv Biobank*. 2015;13(5):307–308.
- Lane JM, Jones SE, Dashti HS, et al. Biological and clinical insights from genetics of insomnia symptoms. *Nat Genet*. 2019;51(3):387–393.
- Jones SE, Lane JM, Wood AR, et al. Genome-wide association analyses of chronotype in 697,828 individuals provides insights into circadian rhythms. *Nat Commun*. 2019;10(1):343.
- Dashti HS, Jones SE, Wood AR, et al. Genome-wide association study identifies genetic loci for self-reported habitual sleep duration supported by accelerometer-derived estimates. *Nat Commun*. 2019;10(1):1100.
- Nebert DW, Liu Z. SLC39A8 gene encoding a metal ion transporter: discovery and bench to bedside. *Hum Genomics*. 2019;13(Suppl 1):51.
- Simpson L, Mukherjee S, Cooper MN, et al. Sex differences in the association of regional fat distribution with the severity of obstructive sleep apnea. *Sleep*. 2010;33(4):467–474.
- Jordan AS, McEvoy RD. Gender differences in sleep apnea: epidemiology, clinical presentation and pathogenic mechanisms. *Sleep Med Rev*. 2003;7(5):377–389.
- Won CHJ, Reid M, Sofer T, et al. Sex differences in obstructive sleep apnea phenotypes, the multi-ethnic study of atherosclerosis. *Sleep*. 2020;43(5):zs274.
- Millman RP, Carlisle CC, McGarvey ST, Eveloff SE, Levinson PD. Body fat distribution and sleep apnea severity in women. *Chest*. 1995;107(2):362–366.
- Liu Y, Zhang X, Lee J, et al. Genome-wide association study of neck circumference identifies sex-specific loci independent of generalized adiposity. *Int J Obes (Lond)*. 2021;45(7):1532–1541.
- Randall JC, Winkler TW, Kutalik Z, et al. Sex-stratified genome-wide association studies including 270,000 individuals show sexual dimorphism in genetic loci for anthropometric traits. *PLoS Genet*. 2013;9(6):e1003500.
- Traglia M, Bseiso D, Gusev A, et al. Genetic mechanisms leading to sex differences across common diseases and anthropometric traits. *Genetics*. 2017;205(2):979–992.
- Khramtsova EA, Davis LK, Stranger BE. The role of sex in the genomics of human complex traits. *Nat Rev Genet*. 2019;20(3):173–190.

- 50 Tramunt B, Smati S, Grandgeorge N, et al. Sex differences in metabolic regulation and diabetes susceptibility. *Diabetologia*. 2020;63(3):453–461.
- 51 Zhang Y, Elgart M, Kurniansyah N, et al. Genetic determinants of cardiometabolic and pulmonary phenotypes and obstructive sleep apnoea in HCHS/SOL. *eBioMedicine*. 2022;84:104288.
- 52 Zhu C, Ming MJ, Cole JM, Kirkpatrick M, Harpak A. Amplification is the primary mode of gene-by-sex interaction in complex human traits. *BioRxiv*. 2022. <https://doi.org/10.1101/2022.05.06.490973>.
- 53 Sofer T. Overcoming the underdiagnosis of OSA to empower genetic association analyses. *Sleep*. 2022;46(3):zsac312.
- 54 Chun S, Akle S, Teodosiadis A, et al. Leveraging pleiotropy to discover and interpret GWAS results for sleep-associated traits. *PLoS Genet*. 2022;18(12):e1010557.
- 55 Veatch OJ, Bauer CR, Keenan BT, et al. Characterization of genetic and phenotypic heterogeneity of obstructive sleep apnea using electronic health records. *BMC Med Genomics*. 2020;13(1):105.
- 56 Duarte J, Pauletto P, Polmann H, et al. Is there an association of genetic polymorphisms of the catechol-O-methyltransferase gene (rs165656 and rs174675) and the 5-hydroxytryptamine receptor 2A gene (rs4941573 and rs6313) with sleep bruxism in individuals with obstructive sleep apnea? *Arch Oral Biol*. 2022;133:105315.
- 57 Wieckiewicz M, Bogunia-Kubik K, Mazur G, et al. Genetic basis of sleep bruxism and sleep apnea-response to a medical puzzle. *Sci Rep*. 2020;10(1):7497.
- 58 Zhang Z, Wang Q, Chen B, Wang Y, Miao Y, Han L. Association study of genetic variations of inflammatory biomarkers with susceptibility and severity of obstructive sleep apnea. *Mol Genet Genomic Med*. 2019;7(8):e801.
- 59 Kerget B, Kerget F, Kahraman ÇY, Aksakal A, Araz Ö. The relationship between NLRP3 rs10159239 and Vaspins rs2236242 gene variants and obstructive sleep apnea. *Ups J Med Sci*. 2021;126.